

Course Code: CS1104		
Name of the Programme: B.Sc(H)		
Name of the Course:		
Semester: I		
Course credit	No. of hours per week (L + T + P) 2:0:2	Total no. of Teaching hours
3	4	60
Course Objectives:		
a) Introduce students to the Linux operating system, its architecture, and various distributions, enabling them to navigate and manage files efficiently.		
b) Equip students with knowledge of shell scripting, including variables, control structures, and functions, to automate tasks and enhance system productivity.		
c) Teach students how to use text processing tools (grep, sed, awk) and regular expressions for efficient data manipulation and extraction.		
d) Enable students to utilize process management techniques and customize their working environment for optimal system performance and usability.		

Syllabus:

Module - 1 Introduction to Linux and Basic Commands	No. of hours
	6
Overview of Linux: Understanding Linux distributions, Basic architecture of Linux. Basic Commands: File management (ls, cp, mv, rm, mkdir, rmdir), Viewing and manipulating file content (cat, more, less, head, tail), Understanding file permissions and ownership (chmod, chown, chgrp).	
Module - 2 Shell Basics and Scripting Fundamentals	No. of hours
	6
Overview of Linux: Introduction to shell types (Bash, Zsh, etc.), Creating and executing simple shell scripts. Working with Variables: Defining and using variables in shell scripts, Environment vs. local variables, Special variables (\$?, \$#, \$*, \$@). Input/Output Operations: Using echo for output, Reading user input with read.	
Module - 3 Control Structures in Shell Scripts	No. of hours
	6
Control Structures: Conditional statements (if, else, elif, case), Looping constructs (for, while). Functions in Shell Scripting: Defining and calling functions, Passing arguments to functions.	

Module - 4 Text Processing Basics	No. of hours
	6
Text Processing Tools: Introduction to filters (grep, sed), Basic text manipulation with awk. Regular Expressions: Basics of regular expressions in Linux, Using regex with tools like grep.	

Module - 5 Process and Customizing the Environment	No. of hours
	6
Process Basics: Process commands (ps), Internal and external commands, Running jobs in the background, nice, at, batch, cron, time commands. Customizing the Environment: System variables, Profile, sty, PWD, Aliases, Command history.	

Course outcomes
a) Explore the architecture of Linux and its distributions to perform effective file, resource, and process management.
b) Apply essential Linux commands for file operations, permission handling, and environment customization.
c) Apply basic shell scripts using variables, control structures, and functions to automate tasks and enhance script functionality.
d) Perform the text processing tools and regular expressions for content manipulation and system scripting.

Text Books	
1	Wale Soyinka (2017). Linux Administration: A Beginner's Guide (6th Edition). McGraw Hill Education. ISBN-13: 978-1259061189.
2	James Arthur (2019). Linux: A Complete Guide to Linux Command Line for Beginners, and How to Get Started with the Linux Operating System! (1st Edition). Ingram Publishing. ISBN-13: 978-1925989717.
3	Sumitabha Das (2017). Linux: Concepts and Applications (4th Edition). McGraw Hill Education. ISBN-13: 978-0070635463.

Reference books	
1	Randal K. Michael (2nd Edition). Mastering Unix Shell Scripting: Bash, Bourne, and Korn Shell Scripting for Programmers, System Administrators, and UNIX Gurus. Wiley. ISBN-13: 978-0470183014. (Kindle Edition)
2	Sunil K. Joseph. Mastering Linux: A Comprehensive Guide to the Operating System. Notion Press. ISBN-13: 979-8891330146.

Lab Programs /Tutorials		30 Hours
1	Perform operations such as creating, copying, moving, and deleting files using command-line commands.	
2	Investigate and modify file permissions for different users and groups.	
3	Use commands to display the content of a file along with line numbers.	
4	Count the number of words, lines, and characters in a specified file.	
5	Perform basic arithmetic operations (addition, subtraction, multiplication, division) based on user input.	
6	Perform various string operations such as concatenation and substring extraction using command-line tools.	
7	Create a script that generates the multiplication table for a specified number.	
8	Remove lines from a file that contain a specific word or phrase.	
9	List all files in the current directory along with their permissions.	
10	Compare the contents of two files to check for differences.	
11	Use commands to display system information such as current date, logged-in users, and system uptime.	
12	End to end System Administration Task Automation Script	
13	End to end IT Support Task Automation Script	